

Chain-of-Thought's Double Edge Is Model-Specific: Prompting versus Scale Across Three Model Families

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Abstract. Practitioners choosing how to spend a limited budget face a recurring question: invest in a larger model, or in better prompting? We study this trade-off in a controlled setting, sweeping three instruction-tuned model families — Qwen2.5 (0.5B–7B), Llama-3 (1B–8B), and Gemma-2 (2B–9B), 4-bit quantization held constant — across four prompting strategies (zero-shot, few-shot, chain-of-thought, and structured JSON output) on three tasks spanning sentiment (SST-2), knowledge (MMLU), and multi-step arithmetic (GSM8K). We quantify prompting as a **parameter-equivalent exchange rate**: how much model scaling a strategy substitutes for. Consistent with recent large-scale evidence that chain-of-thought helps mainly on reasoning and barely on knowledge (Sprague et al., 2024), we confirm in this controlled small-model setting that the exchange rate is **largest on reasoning and smallest on knowledge**. On GSM8K, chain-of-thought (CoT) at a given size exceeds zero-shot accuracy at a much larger size — a Qwen-1.5B with CoT (55.7%) outperforms a Qwen-7B without it (14.3%), and a Llama-3B with CoT (74.7%) beats a Llama-8B without it (22.7%). On MMLU, no strategy improves on zero-shot by more than a few points; accuracy is governed almost entirely by parameter count. These effects replicate across all three families. Our central finding is that CoT's effect on knowledge is model-specific, and the three families show three distinct patterns: in Qwen2.5, CoT *degrades* multiple-choice knowledge QA increasingly with scale (−7 to −17pp); in Gemma-2 it harms the small model but the harm fades with scale (−15pp at 2B to −1pp at 9B); and in Llama-3 it is neutral-to-helpful. Whether CoT helps or hurts knowledge — and how that changes with scale — depends on the model family. Robustness checks (two random seeds, a continuous numeric-presence metric, and re-extraction from full outputs) confirm these effects are not scoring artifacts. All experiments run on a single 16GB laptop; code and prompts are released at github.com/ssamalsamir/cot-model-specificity.

1. Introduction

Deploying a language model means choosing both *which model* and *how to prompt it*, and these choices compete for the same budget: a larger model costs compute, memory, and latency, while prompt engineering is nearly free but bounded in effect. Prior work establishes the two levers separately — that few-shot ability emerges with scale (Brown et al., 2020), that chain-of-thought helps larger models reason (Wei et al., 2022b; Kojima et al., 2022), and that accuracy follows smooth scaling laws (Kaplan et al., 2020; Hoffmann et al., 2022) — but rarely quantifies how the two trade off against each other on a single, controlled model family.

We ask two questions. **RQ1 (exchange rate):** how much model scaling does each prompting strategy substitute for, and does this depend on task type? **RQ2 (CoT's cost):** does the strategy that helps reasoning carry over to other task types, or can it hurt?

Recent large-scale evidence (Sprague et al., 2024) shows chain-of-thought helps mainly on math and symbolic reasoning and adds little on knowledge tasks. We first **confirm this in a controlled, same-family, sub-7B setting**: a good prompt buys several model tiers on GSM8K yet almost nothing on MMLU, where facts must live in the parameters. Our central *new* contribution is then to show that one such prompting effect is **model-family-specific** — across three families, chain-of-thought ranges from actively harming knowledge questions (Qwen2.5, Gemma-2) to helping them (Llama-3), and the direction of the harm's scale-trend itself differs by family (§4.3).

Contributions. (1) **Our primary contribution:** chain-of-thought's effect on multiple-choice knowledge QA is *model-family-specific*, with three distinct patterns — CoT degrades MMLU increasingly with scale in Qwen2.5 (−7 to −17pp), harms the small model but fades with scale in Gemma-2 (−15pp → −1pp), and is neutral-to-helpful in Llama-3 — verified against extraction and metric artifacts. (2) A controlled, same-family confirmation, in the under-studied sub-7B locally-quantized regime, of the recent result (Sprague et al., 2024) that prompting (chiefly CoT) substitutes for scale on reasoning but not on knowledge, which we quantify as a prompt–parameter exchange rate. (3) Statistical rigor uncommon at this scale: three

model families, up to two seeds ($n = 300\text{--}600$ per condition), Wilson confidence intervals, and Holm-adjusted McNemar tests. (4) A reproducible single-laptop harness, with code and prompts released.

2. Related Work

Prompting and scale. In-context few-shot learning emerges with model size (Brown et al., 2020); chain-of-thought prompting improves reasoning primarily in larger models (Wei et al., 2022b; Kojima et al., 2022). The interaction between few-shot benefit and model size has been studied directly (San Joaquin & Haroen, 2022), and recent work shows small models struggle to exploit long reasoning chains (Li et al., 2025). Most directly, Sprague et al. (2024) show, across many models and datasets, that CoT's gains concentrate in math and symbolic reasoning and are minimal on knowledge and language-understanding tasks; we confirm this in a controlled same-family, sub-7B setting and extend it by showing one such effect can be model-family-specific. None of these reports a quantitative prompt-as-parameters exchange rate on a controlled same-family ladder, nor the family-specific knowledge-QA degradation we observe.

Scaling and emergence. Performance follows power-law scaling (Kaplan et al., 2020; Hoffmann et al., 2022). Some abilities appear abruptly with scale (Wei et al., 2022a), though this "emergence" may partly be an artifact of discontinuous metrics (Schaeffer et al., 2023) — a concern we address directly in §4.4.

Format and prompt sensitivity. Enforced output formats can degrade reasoning (Tam et al., 2024); meaning-preserving format changes swing accuracy substantially and the sensitivity persists across scale (Sclar et al., 2023); few-shot example ordering alone can dominate performance (Lu et al., 2022). GSM8K itself may reflect brittle pattern-matching (Mirzadeh et al., 2024). Our closest points of comparison are Tam et al. (2024), who study format restrictions but across a heterogeneous set of models rather than a controlled size ladder, and San Joaquin & Haroen (2022); neither frames results as an exchange rate nor contrasts knowledge against multi-step reasoning as we do.

3. Methods

Models. Three instruction-tuned families swept across parameter count with 4-bit quantization held constant, so size is the only within-family variable: Qwen2.5-Instruct at 0.5B / 1.5B / 3B / 7B, Llama-3 at 1B / 3B / 8B, and Gemma-2 at 2B / 9B. We run locally via MLX on an Apple M4 (16GB), one model resident at a time.

Tasks. SST-2 (binary sentiment), MMLU (4-way multiple-choice knowledge, stratified across 57 subjects), GSM8K (grade-school multi-step arithmetic). $n = 300$ examples per condition. Qwen2.5 and Llama-3 were evaluated at two seeds (42 and 43; pooled $n = 600$); Gemma-2 was evaluated at one seed (42; $n = 300$). Confidence intervals are reported in Appendix C.

Strategies. Zero-shot; few-shot (fixed hand-written exemplars); CoT (reasoning followed by a parsed final answer); structured (JSON-only output). Full templates in Appendix A.

Decoding and scoring. Greedy decoding (temperature 0), deterministic. Task-specific answer extraction: numeric match for GSM8K, exact label/letter match otherwise.

Statistics. Accuracy with 95% Wilson score intervals ($\approx \pm 5\text{--}6\text{pp}$ at $n = 300$); paired McNemar tests (continuity-corrected, Holm-adjusted) for within-model strategy comparisons. We define the **exchange rate** operationally: prompting at size S "matches" scaling if the best-prompt accuracy at S meets or exceeds zero-shot accuracy at the next-larger size.

Robustness. To address the concern that exact-match exaggerates discontinuities (Schaeffer et al., 2023), for GSM8K we additionally record *numeric-present* (whether the correct value appears anywhere in the output) and the relative error of the committed answer; and we re-extract answers from full (untruncated) outputs (§4.4).

4. Results

4.1 Main results

Figure 1. Accuracy vs. model scale, by prompting strategy

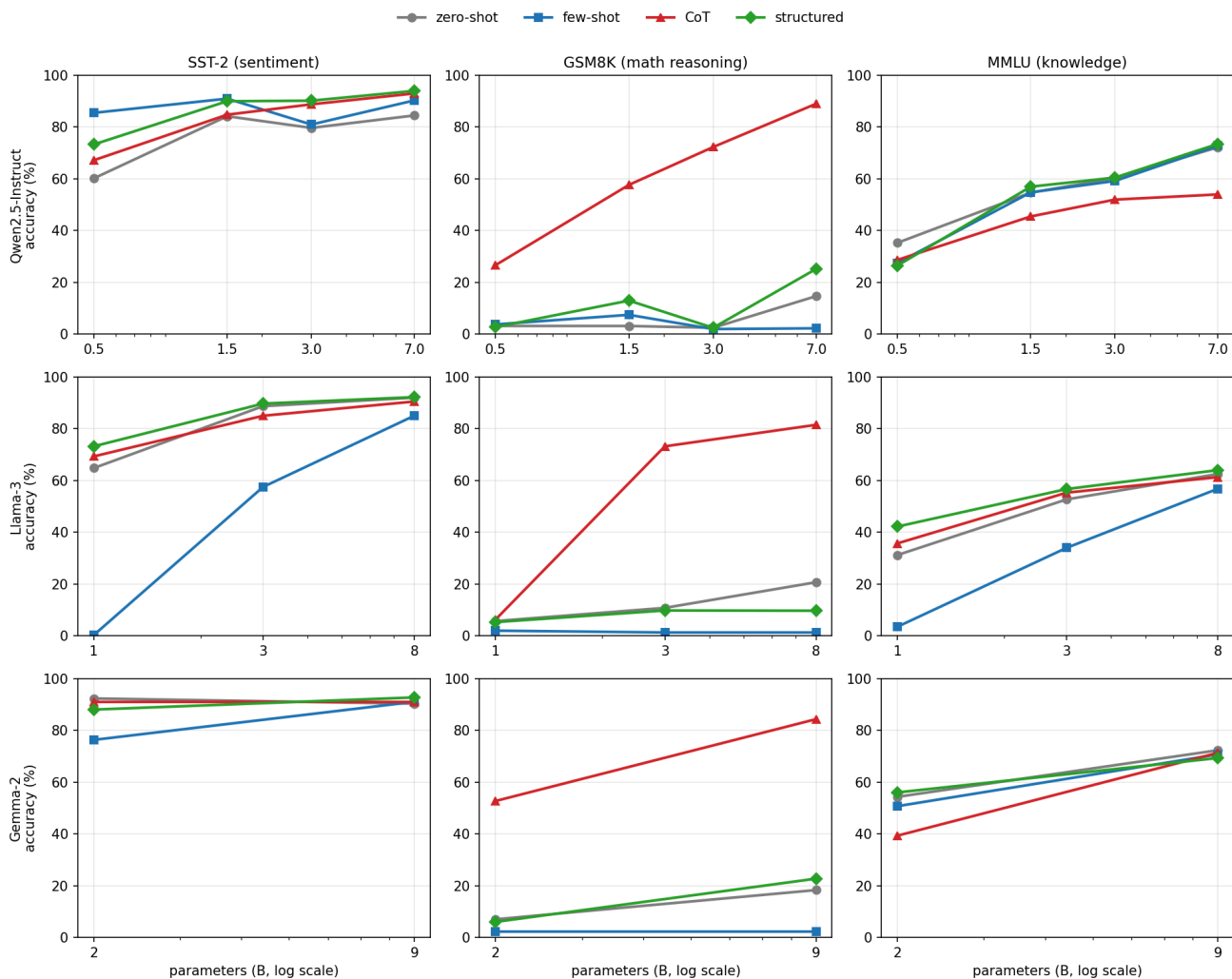


Figure 1. Accuracy vs. model scale (log axis) for each prompting strategy, by family (rows: Qwen2.5, Llama-3, Gemma-2) and task (columns). On GSM8K, the CoT curve (red) separates sharply from all others in every family; on MMLU the strategies bunch together and rise mainly with scale; on SST-2 the spread is moderate. Note Llama-3's erratic few-shot curve (blue).

Task	Model	zero-shot	few-shot	CoT	structured
SST-2	Qwen-0.5B	62.0	85.3	69.7	72.0
	Qwen-1.5B	86.7	92.7	84.3	91.0
	Qwen-3B	80.7	81.7	89.0	90.7
	Qwen-7B	87.3	91.0	94.7	95.3
MMLU	Qwen-0.5B	32.7	26.3	25.3	25.0
	Qwen-1.5B	57.3	55.3	47.3	60.0
	Qwen-3B	59.7	59.7	49.3	60.0
	Qwen-7B	72.3	74.0	55.3	73.7
GSM8K	Qwen-0.5B	4.3	4.0	25.3	2.3
	Qwen-1.5B	2.3	8.3	55.7	14.3
	Qwen-3B	2.3	1.7	70.7	2.7
	Qwen-7B	14.3	2.7	88.0	24.0

Table 1. Qwen2.5 accuracy (%). Green = best strategy for that row; red = CoT underperforms zero-shot.

Task	Model	zero-shot	few-shot	CoT	structured
SST-2	Llama-1B	67.3	0.7	71.0	72.0
	Llama-3B	89.3	60.0	86.0	90.0
	Llama-8B	92.0	85.0	90.3	91.7
MMLU	Llama-1B	31.3	4.0	37.3	38.3
	Llama-3B	52.3	35.0	56.3	57.3
	Llama-8B	60.7	55.0	58.7	62.3
GSM8K	Llama-1B	6.3	1.3	7.3	6.7
	Llama-3B	10.0	1.3	74.7	10.7
	Llama-8B	22.7	1.0	79.0	9.7

Table 2. Llama-3 accuracy (%). Note few-shot's collapse (red) and that CoT does *not* underperform zero-shot on MMLU here — the key cross-family divergence.

Task	Model	zero-shot	few-shot	CoT	structured
SST-2	Gemma-2B	92.3	76.3	91.0	88.0
	Gemma-9B	90.3	91.0	91.0	92.7
MMLU	Gemma-2B	54.3	50.7	39.3	56.0
	Gemma-9B	72.3	70.7	71.3	69.3
GSM8K	Gemma-2B	7.0	2.3	52.7	6.0
	Gemma-9B	18.3	2.3	84.3	22.7

Table 3. Gemma-2 accuracy (% , seed 42, n = 300). On MMLU, CoT underperforms zero-shot (red) by −15pp at 2B but only −1pp at 9B — the harm *fades* with scale, unlike Qwen (worsens) or Llama (neutral).

4.2 The exchange rate is large on reasoning, near-zero on knowledge (RQ1)

Figure 3. The prompt-parameter exchange rate on GSM8K: a small model with CoT beats a much larger one without it

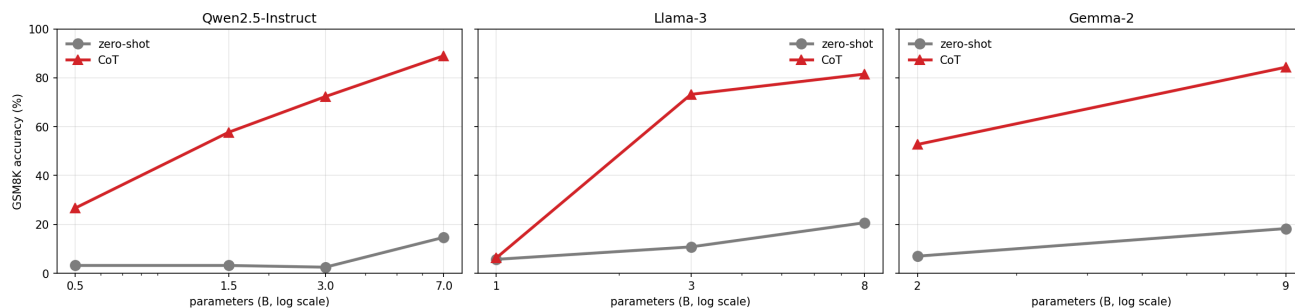


Figure 3. Zero-shot vs. CoT accuracy on GSM8K. In all three families, CoT at a small size sits far above zero-shot at a much larger size — e.g. Qwen-1.5B + CoT (55.7%) exceeds Qwen-7B zero-shot (14.3%); Llama-3B + CoT (74.7%) exceeds Llama-8B zero-shot (22.7%); Gemma-2B + CoT (52.7%) exceeds Gemma-9B zero-shot (18.3%).

GSM8K — prompting » scaling. CoT at every size exceeds zero-shot at the next size up. A 1.5B Qwen with CoT outperforms a 7B Qwen without it by 41pp — prompting worth roughly a $5\times$ parameter increase — and the pattern repeats in Llama and Gemma (Gemma-2B + CoT 52.7% beats Gemma-9B zero-shot 18.3%). Crucially this is CoT-specific: few-shot and structured remain near the floor, because answer-only or format-constrained prompts suppress the reasoning the task requires. (Llama-1B is an instructive floor: too weak to reason even with CoT.)

MMLU — scaling is irreplaceable. The best strategy improves on zero-shot by at most a few points (≤ 2.7 pp in Qwen, ≤ 7 pp in Llama, ≤ 1.7 pp in Gemma). Accuracy rises with parameters regardless of prompt. Knowledge either resides in the weights or it does not. This reasoning-versus-knowledge split confirms, in a controlled small-model setting, the broad pattern reported by Sprague et al. (2024).

SST-2 — prompting helps moderately, with best-strategy gains of 6–23pp, often equivalent to about one model tier.

4.3 CoT is double-edged — but the cost is model-specific (RQ2)

Figure 2. Chain-of-thought helps reasoning everywhere, but its effect on knowledge (MMLU) is model-specific

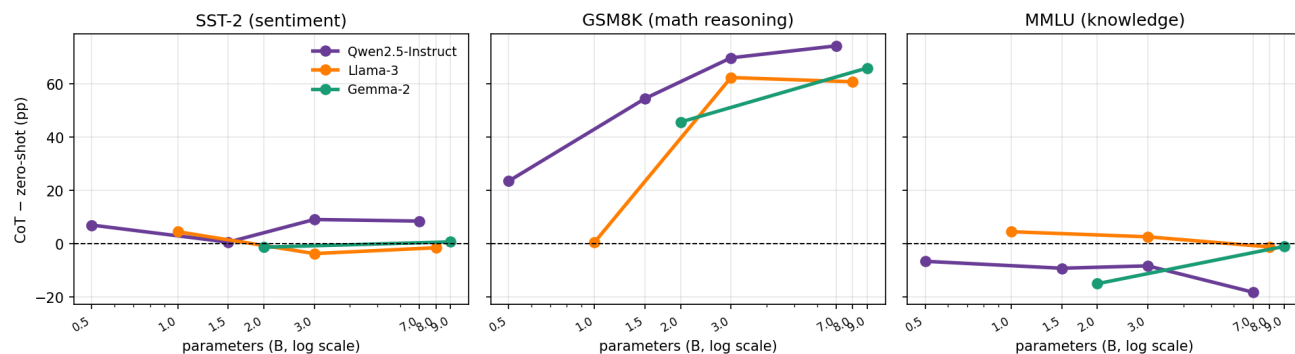


Figure 2. CoT minus zero-shot accuracy (pp) vs. model size. On GSM8K all three families are large and positive. On MMLU they diverge into three patterns: Qwen (purple) drops increasingly below zero (worst at 7B, -17pp); Gemma (green) starts low (-15pp at 2B) but recovers toward zero by 9B; Llama (orange) stays neutral-to-positive. On SST-2 all are small.

In Qwen2.5, the same CoT that dominates GSM8K *degrades* MMLU at every size — by 7.4pp (0.5B), 10.0pp (1.5B), 10.4pp (3B), and 17.0pp (7B) — with harm growing in scale. We interpret this as over-reasoning on multiple-choice items: free-form reasoning creates opportunities to argue away the correct option (cf. Tam et al., 2024). Notably, a recent meta-analysis finds

CoT roughly *neutral* on MMLU on average (Sprague et al., 2024), so an active, scale-increasing degradation within a single family is a stronger and more specific effect than previously reported.

The three families show three distinct patterns. In Llama-3, CoT on MMLU is neutral-to-helpful (+6.0pp at 1B, +4.0pp at 3B, -2.0pp at 8B). In Gemma-2, CoT harms the small model (-15.0pp at 2B) but the harm *fades* with scale to near-zero (-1.0pp at 9B) — the opposite scale-trend to Qwen, where the harm *grows*. So across three families CoT’s effect on knowledge QA ranges from helpful (Llama) to harmful-and-worsening (Qwen) to harmful-but-improving (Gemma). We therefore report it as a *model-family-dependent phenomenon, not a universal law*, while the broader claim — that prompting buys little on knowledge regardless of strategy (§4.2) — holds in all three families. This divergence is exactly what a single-family study would have over-generalized.

Conversely, structured (JSON) output is *not* a universal penalty in any family — it is best-in-class on SST-2 (Qwen-7B: 95.3; Llama-8B: 91.7; Gemma-9B: 92.7) and competitive on MMLU, and only clearly harmful on GSM8K, refining the "format tax" of Tam et al. (2024) to a reasoning-specific effect. We also observe that few-shot is markedly less reliable in Llama-3 than Qwen2.5 (e.g. GSM8K $\approx 1\%$, SST-2-1B 0.7%), consistent with strong family-specific sensitivity to exemplar formatting (Sclar et al., 2023; Lu et al., 2022).

4.4 The effects are not scoring artifacts

Continuous metric (GSM8K). Numeric-present tracks exact-match closely rather than diverging (Qwen-0.5B CoT 25.3 / 39.0; Qwen-7B CoT 88.0 / 91.0), and the median relative error of the committed answer is 0.000 for CoT at 1.5B and above. The low small-model scores are not an artifact of a discontinuous metric (Schaeffer et al., 2023).

Extraction audit (MMLU CoT). Re-extracting answers from full outputs, the extraction-failure rate is only 2–4%; scoring only cleanly-parsed answers, Qwen CoT still trails zero-shot by 6–16pp at every size. The CoT-hurts-knowledge effect in Qwen is real, not a parsing failure.

5. Discussion

Our central observation is that a single prompting technique can help one model family and hurt another on the same task: chain-of-thought lifts all three families on math, but on multiple-choice knowledge it degrades Qwen (increasingly with scale), harms then largely spares Gemma (the penalty fades from -15pp at 2B to -1pp at 9B), and leaves Llama unharmed. Across three families we see three distinct behaviours, and the *scale-trend* itself differs — the harm grows for Qwen but shrinks for Gemma. Because each pattern is consistent across sizes within its family, it is unlikely to be noise; still, three families is a small sample, so we treat the specific taxonomy as an observation to be tested more broadly, not a settled law. A plausible mechanism is over-reasoning: free-form deliberation gives a model room to talk itself out of a correct option, and families differ both in how susceptible they are and in whether that susceptibility grows or shrinks with scale.

The broader prompting-versus-scale pattern we confirm — prompting substitutes for scale on reasoning but barely on knowledge (Sprague et al., 2024) — has a clean intuition: reasoning tasks hold latent capability a prompt can elicit, whereas a fact is either in the weights or it is not, so only scale moves it. The practical corollary: under a fixed budget, spend on prompting for reasoning workloads and on parameters for knowledge workloads — and do not assume chain-of-thought is safe on knowledge questions, because whether it helps depends on the model.

6. Limitations

- **Three families, and Gemma at one seed.** Each family’s CoT-on-knowledge pattern is consistent across its sizes, but three families — with Gemma at only two sizes and a single seed ($n = 300$, versus $n = 600$ for Qwen and Llama) — cannot establish the three-pattern taxonomy as general; more families, sizes, and seeds (and a Gemma second seed) are needed to confirm it generalizes.

- **4-bit quantization throughout.** Held constant, but it may interact with size or family, so a quantization contribution to the cross-family difference cannot be fully ruled out; an unquantized control would strengthen the claim.
- **Seeds.** For Qwen and Llama, accuracies across seeds 42 and 43 differed by $\sim 2.3\text{pp}$ on average (max 7.7pp), within the Wilson intervals, and every effect replicated — but this is not a large-sample variance estimate, and Gemma was run at a single seed.
- **Greedy decoding only** — no self-consistency or temperature sweep, either of which could shift CoT.
- **Fixed hand-written exemplars.** Prompt-order and format sensitivity were not ablated (Lu et al., 2022; Sclar et al., 2023); Llama's erratic few-shot is likely an instance of this.
- **Scope.** Three benchmarks, English-only. GSM8K may probe brittle pattern-matching rather than reasoning (Mirzadeh et al., 2024), so the "reasoning" reading is a caveat.
- **Metric.** The Schaeffer (2023) concern is mitigated by the numeric-present metric, but exact-match remains the headline; we report both. The "exchange rate" is a descriptive operational metric, not a law.

7. Conclusion

In a controlled three-family sweep, we find that chain-of-thought's effect on knowledge questions is model-dependent, with three distinct patterns — actively harmful and worsening with scale for Qwen2.5, harmful at small scale but fading for Gemma-2, and neutral-to-helpful for Llama-3 — a divergence a single-family study would have missed. This sits within a broader pattern, consistent with prior large-scale evidence (Sprague et al., 2024), that prompting substitutes for scale on reasoning but barely on knowledge. The lever you should pull depends on the task and the model; whether a prompt that helps one model helps another is not safe to assume.

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Appendix A. Prompt templates

Each prompt below is wrapped in the model's chat template (`apply_chat_template` , `add_generation_prompt=True`) before generation. `{...}` denotes the inserted example.

A.1 SST-2 (sentiment)

zero-shot: Classify the sentiment of the following sentence as exactly
'positive' or 'negative'.

Sentence: {sentence}
Sentiment:

few-shot: Classify the sentiment as 'positive' or 'negative'.
[3 exemplars: ("The film is a masterpiece of visual storytelling." -> positive),
("I fell asleep twice. Complete waste of time." -> negative),
("A quietly moving portrait of grief and resilience." -> positive)]

Sentence: {sentence}
Sentiment:

CoT: Identify the key sentiment signals in the sentence, reason through them,
then classify as 'positive' or 'negative'. End your response with
'Final answer: positive' or 'Final answer: negative'.

Sentence: {sentence}
Reasoning:

structured: Classify sentiment. Respond ONLY with valid JSON:
{"sentiment": "positive"} or {"sentiment": "negative"}.

Sentence: {sentence}
JSON:

A.2 GSM8K (arithmetic)

zero-shot: Solve the following math problem. Give only the final number as your
answer.

Problem: {question}
Answer:

few-shot: 2 worked examples shown as Problem / Answer (final number only),
then Problem: {question}
Answer:

CoT: 2 worked examples shown as Problem / Reasoning / Answer, then
"Solve ... step by step. End with 'Answer: <number>'."

Problem: {question}
Reasoning:

structured: Solve this math problem. Respond ONLY with valid JSON:

```
{"answer": "<number>"}
```

Problem: {question}

JSON:

A.3 MMLU (knowledge, 4-way MC)

zero-shot: Answer the following multiple choice question. Respond with only the letter A, B, C, or D.

Question: {q}

{A..D options}

Answer:

few-shot: 4 exemplars (answers spread across A/B/C/D to avoid position bias), then Question: {q}

{options}

Answer:

CoT: "Think through each option carefully, then end your response with 'Final answer: A' (or B, C, D)."

Question: {q}

{options}

Reasoning:

structured: Respond ONLY with valid JSON: {"answer": "A"} (or B, C, D).

Question: {q}

{options}

JSON:

Appendix B. Answer-extraction rules

Greedy decoding (temperature 0); answers parsed from raw model text as follows.

- **SST-2**. structured: parse JSON `sentiment` field. CoT: regex `final answer[:\s]+(positive|negative)` . Otherwise: scan the last 60 characters, then the whole output, for "positive"/"negative".
- **GSM8K**. structured: parse JSON `answer` . Otherwise: first match of `answer[:\s]+$(number)` , then a trailing `= $number` , then the last number in the text (excluding bare years 2020-2026). Scored by numeric equality (tolerance 0.01).
- **MMLU**. structured: parse JSON `answer` . CoT: regex `final answer[:\s]+([A-D])` . Otherwise: a lone A-D letter on the last non-empty line, then the last A-D letter anywhere. Scored by exact letter match.

Extraction failures count as incorrect. An audit of MMLU-CoT on full (untruncated) outputs found a 2-4% extraction-failure rate, with the reported effects unchanged (§4.4).

Appendix C. Full per-condition results (95% Wilson CIs & McNemar tests)

Qwen2.5 and Llama-3 are pooled over seeds 42 and 43 (n = 600 per condition); Gemma-2 uses seed 42 (n = 300). McNemar tests are continuity-corrected and Holm-adjusted within each model×task, comparing each strategy to zero-shot on the same

items.

C.1 Accuracy and 95% Wilson CI

Family	Model	Task	Strategy	n	Acc (%)	95% CI
Qwen2.5-Instruct	qwen-0.5b	sst2	zero-shot	600	60.2	[56.2, 64.0]
Qwen2.5-Instruct	qwen-0.5b	sst2	few-shot	600	85.5	[82.5, 88.1]
Qwen2.5-Instruct	qwen-0.5b	sst2	CoT	600	67.2	[63.3, 70.8]
Qwen2.5-Instruct	qwen-0.5b	sst2	structured	600	73.3	[69.7, 76.7]
Qwen2.5-Instruct	qwen-0.5b	gsm8k	zero-shot	600	3.2	[2.0, 4.9]
Qwen2.5-Instruct	qwen-0.5b	gsm8k	few-shot	600	3.8	[2.6, 5.7]
Qwen2.5-Instruct	qwen-0.5b	gsm8k	CoT	600	26.7	[23.3, 30.3]
Qwen2.5-Instruct	qwen-0.5b	gsm8k	structured	600	2.8	[1.8, 4.5]
Qwen2.5-Instruct	qwen-0.5b	mmlu	zero-shot	600	35.3	[31.6, 39.2]
Qwen2.5-Instruct	qwen-0.5b	mmlu	few-shot	600	27.5	[24.1, 31.2]
Qwen2.5-Instruct	qwen-0.5b	mmlu	CoT	600	28.7	[25.2, 32.4]
Qwen2.5-Instruct	qwen-0.5b	mmlu	structured	600	26.5	[23.1, 30.2]
Qwen2.5-Instruct	qwen-1.5b	sst2	zero-shot	600	84.2	[81.0, 86.9]
Qwen2.5-Instruct	qwen-1.5b	sst2	few-shot	600	91.0	[88.4, 93.0]
Qwen2.5-Instruct	qwen-1.5b	sst2	CoT	600	84.8	[81.7, 87.5]
Qwen2.5-Instruct	qwen-1.5b	sst2	structured	600	90.0	[87.3, 92.2]
Qwen2.5-Instruct	qwen-1.5b	gsm8k	zero-shot	600	3.2	[2.0, 4.9]
Qwen2.5-Instruct	qwen-1.5b	gsm8k	few-shot	600	7.5	[5.7, 9.9]
Qwen2.5-Instruct	qwen-1.5b	gsm8k	CoT	600	57.7	[53.7, 61.6]
Qwen2.5-Instruct	qwen-1.5b	gsm8k	structured	600	13.0	[10.5, 15.9]
Qwen2.5-Instruct	qwen-1.5b	mmlu	zero-shot	600	54.7	[50.7, 58.6]
Qwen2.5-Instruct	qwen-1.5b	mmlu	few-shot	600	54.8	[50.8, 58.8]
Qwen2.5-Instruct	qwen-1.5b	mmlu	CoT	600	45.5	[41.6, 49.5]
Qwen2.5-Instruct	qwen-1.5b	mmlu	structured	600	57.0	[53.0, 60.9]
Qwen2.5-Instruct	qwen-3b	sst2	zero-shot	600	79.7	[76.3, 82.7]
Qwen2.5-Instruct	qwen-3b	sst2	few-shot	600	81.0	[77.7, 83.9]
Qwen2.5-Instruct	qwen-3b	sst2	CoT	600	88.8	[86.1, 91.1]
Qwen2.5-Instruct	qwen-3b	sst2	structured	600	90.2	[87.5, 92.3]
Qwen2.5-Instruct	qwen-3b	gsm8k	zero-shot	600	2.5	[1.5, 4.1]
Qwen2.5-Instruct	qwen-3b	gsm8k	few-shot	600	2.0	[1.1, 3.5]
Qwen2.5-Instruct	qwen-3b	gsm8k	CoT	600	72.3	[68.6, 75.8]
Qwen2.5-Instruct	qwen-3b	gsm8k	structured	600	2.5	[1.5, 4.1]
Qwen2.5-Instruct	qwen-3b	mmlu	zero-shot	600	60.3	[56.4, 64.2]
Qwen2.5-Instruct	qwen-3b	mmlu	few-shot	600	59.2	[55.2, 63.0]

Qwen2.5-Instruct	qwen-3b	mmlu	CoT	600	52.0	[48.0, 56.0]
Qwen2.5-Instruct	qwen-3b	mmlu	structured	600	60.5	[56.5, 64.3]
Qwen2.5-Instruct	qwen-7b	sst2	zero-shot	600	84.5	[81.4, 87.2]
Qwen2.5-Instruct	qwen-7b	sst2	few-shot	600	90.3	[87.7, 92.4]
Qwen2.5-Instruct	qwen-7b	sst2	CoT	600	93.0	[90.7, 94.8]
Qwen2.5-Instruct	qwen-7b	sst2	structured	600	94.0	[91.8, 95.6]
Qwen2.5-Instruct	qwen-7b	gsm8k	zero-shot	600	14.7	[12.1, 17.7]
Qwen2.5-Instruct	qwen-7b	gsm8k	few-shot	600	2.3	[1.4, 3.9]
Qwen2.5-Instruct	qwen-7b	gsm8k	CoT	600	89.0	[86.2, 91.3]
Qwen2.5-Instruct	qwen-7b	gsm8k	structured	600	25.2	[21.9, 28.8]
Qwen2.5-Instruct	qwen-7b	mmlu	zero-shot	600	72.2	[68.4, 75.6]
Qwen2.5-Instruct	qwen-7b	mmlu	few-shot	600	72.8	[69.1, 76.2]
Qwen2.5-Instruct	qwen-7b	mmlu	CoT	600	54.0	[50.0, 57.9]
Qwen2.5-Instruct	qwen-7b	mmlu	structured	600	73.5	[69.8, 76.9]
Llama-3	llama-1b	sst2	zero-shot	600	64.8	[60.9, 68.5]
Llama-3	llama-1b	sst2	few-shot	600	0.3	[0.1, 1.2]
Llama-3	llama-1b	sst2	CoT	600	69.3	[65.5, 72.9]
Llama-3	llama-1b	sst2	structured	600	73.2	[69.5, 76.6]
Llama-3	llama-1b	gsm8k	zero-shot	600	5.7	[4.1, 7.8]
Llama-3	llama-1b	gsm8k	few-shot	600	2.0	[1.1, 3.5]
Llama-3	llama-1b	gsm8k	CoT	600	6.3	[4.6, 8.6]
Llama-3	llama-1b	gsm8k	structured	600	5.3	[3.8, 7.4]
Llama-3	llama-1b	mmlu	zero-shot	600	31.2	[27.6, 35.0]
Llama-3	llama-1b	mmlu	few-shot	600	3.5	[2.3, 5.3]
Llama-3	llama-1b	mmlu	CoT	600	35.7	[31.9, 39.6]
Llama-3	llama-1b	mmlu	structured	600	42.2	[38.3, 46.2]
Llama-3	llama-3b	sst2	zero-shot	600	88.7	[85.9, 91.0]
Llama-3	llama-3b	sst2	few-shot	600	57.5	[53.5, 61.4]
Llama-3	llama-3b	sst2	CoT	600	85.0	[81.9, 87.6]
Llama-3	llama-3b	sst2	structured	600	89.7	[87.0, 91.9]
Llama-3	llama-3b	gsm8k	zero-shot	600	10.8	[8.6, 13.6]
Llama-3	llama-3b	gsm8k	few-shot	600	1.3	[0.7, 2.6]
Llama-3	llama-3b	gsm8k	CoT	600	73.2	[69.5, 76.6]
Llama-3	llama-3b	gsm8k	structured	600	9.8	[7.7, 12.5]
Llama-3	llama-3b	mmlu	zero-shot	600	52.7	[48.7, 56.6]
Llama-3	llama-3b	mmlu	few-shot	600	34.0	[30.3, 37.9]
Llama-3	llama-3b	mmlu	CoT	600	55.3	[51.3, 59.3]

Llama-3	llama-3b	mmlu	structured	600	56.7	[52.7, 60.6]
Llama-3	llama-8b	sst2	zero-shot	600	92.0	[89.6, 93.9]
Llama-3	llama-8b	sst2	few-shot	600	85.0	[81.9, 87.6]
Llama-3	llama-8b	sst2	CoT	600	90.5	[87.9, 92.6]
Llama-3	llama-8b	sst2	structured	600	92.2	[89.7, 94.1]
Llama-3	llama-8b	gsm8k	zero-shot	600	20.7	[17.6, 24.1]
Llama-3	llama-8b	gsm8k	few-shot	600	1.3	[0.7, 2.6]
Llama-3	llama-8b	gsm8k	CoT	600	81.5	[78.2, 84.4]
Llama-3	llama-8b	gsm8k	structured	600	9.7	[7.6, 12.3]
Llama-3	llama-8b	mmlu	zero-shot	600	62.5	[58.6, 66.3]
Llama-3	llama-8b	mmlu	few-shot	600	56.8	[52.8, 60.7]
Llama-3	llama-8b	mmlu	CoT	600	61.3	[57.4, 65.1]
Llama-3	llama-8b	mmlu	structured	600	64.0	[60.1, 67.7]
Gemma-2	gemma-2b	sst2	zero-shot	300	92.3	[88.8, 94.8]
Gemma-2	gemma-2b	sst2	few-shot	300	76.3	[71.2, 80.8]
Gemma-2	gemma-2b	sst2	CoT	300	91.0	[87.2, 93.7]
Gemma-2	gemma-2b	sst2	structured	300	88.0	[83.8, 91.2]
Gemma-2	gemma-2b	gsm8k	zero-shot	300	7.0	[4.6, 10.5]
Gemma-2	gemma-2b	gsm8k	few-shot	300	2.3	[1.1, 4.7]
Gemma-2	gemma-2b	gsm8k	CoT	300	52.7	[47.0, 58.2]
Gemma-2	gemma-2b	gsm8k	structured	300	6.0	[3.8, 9.3]
Gemma-2	gemma-2b	mmlu	zero-shot	300	54.3	[48.7, 59.9]
Gemma-2	gemma-2b	mmlu	few-shot	300	50.7	[45.0, 56.3]
Gemma-2	gemma-2b	mmlu	CoT	300	39.3	[34.0, 45.0]
Gemma-2	gemma-2b	mmlu	structured	300	56.0	[50.3, 61.5]
Gemma-2	gemma-9b	sst2	zero-shot	300	90.3	[86.5, 93.2]
Gemma-2	gemma-9b	sst2	few-shot	300	91.0	[87.2, 93.7]
Gemma-2	gemma-9b	sst2	CoT	300	91.0	[87.2, 93.7]
Gemma-2	gemma-9b	sst2	structured	300	92.7	[89.1, 95.1]
Gemma-2	gemma-9b	gsm8k	zero-shot	300	18.3	[14.4, 23.1]
Gemma-2	gemma-9b	gsm8k	few-shot	300	2.3	[1.1, 4.7]
Gemma-2	gemma-9b	gsm8k	CoT	300	84.3	[79.8, 88.0]
Gemma-2	gemma-9b	gsm8k	structured	300	22.7	[18.3, 27.7]
Gemma-2	gemma-9b	mmlu	zero-shot	300	72.3	[67.0, 77.1]
Gemma-2	gemma-9b	mmlu	few-shot	300	70.7	[65.3, 75.5]
Gemma-2	gemma-9b	mmlu	CoT	300	71.3	[66.0, 76.2]
Gemma-2	gemma-9b	mmlu	structured	300	69.3	[63.9, 74.3]

C.2 McNemar tests vs. zero-shot (Holm-adjusted)

Family	Model	Task	Comparison	p (adj)	
Qwen2.5-Instruct	qwen-0.5b	sst2	zero-shot vs few-shot	0.0000	***
Qwen2.5-Instruct	qwen-0.5b	sst2	zero-shot vs CoT	0.0006	***
Qwen2.5-Instruct	qwen-0.5b	sst2	zero-shot vs structured	0.0000	***
Qwen2.5-Instruct	qwen-0.5b	gsm8k	zero-shot vs few-shot	1.0000	ns
Qwen2.5-Instruct	qwen-0.5b	gsm8k	zero-shot vs CoT	0.0000	***
Qwen2.5-Instruct	qwen-0.5b	gsm8k	zero-shot vs structured	1.0000	ns
Qwen2.5-Instruct	qwen-0.5b	mmlu	zero-shot vs few-shot	0.0002	***
Qwen2.5-Instruct	qwen-0.5b	mmlu	zero-shot vs CoT	0.0015	**
Qwen2.5-Instruct	qwen-0.5b	mmlu	zero-shot vs structured	0.0000	***
Qwen2.5-Instruct	qwen-1.5b	sst2	zero-shot vs few-shot	0.0000	***
Qwen2.5-Instruct	qwen-1.5b	sst2	zero-shot vs CoT	0.7373	ns
Qwen2.5-Instruct	qwen-1.5b	sst2	zero-shot vs structured	0.0002	***
Qwen2.5-Instruct	qwen-1.5b	gsm8k	zero-shot vs few-shot	0.0010	**
Qwen2.5-Instruct	qwen-1.5b	gsm8k	zero-shot vs CoT	0.0000	***
Qwen2.5-Instruct	qwen-1.5b	gsm8k	zero-shot vs structured	0.0000	***
Qwen2.5-Instruct	qwen-1.5b	mmlu	zero-shot vs few-shot	1.0000	ns
Qwen2.5-Instruct	qwen-1.5b	mmlu	zero-shot vs CoT	0.0004	***
Qwen2.5-Instruct	qwen-1.5b	mmlu	zero-shot vs structured	0.2922	ns
Qwen2.5-Instruct	qwen-3b	sst2	zero-shot vs few-shot	0.5045	ns
Qwen2.5-Instruct	qwen-3b	sst2	zero-shot vs CoT	0.0000	***
Qwen2.5-Instruct	qwen-3b	sst2	zero-shot vs structured	0.0000	***
Qwen2.5-Instruct	qwen-3b	gsm8k	zero-shot vs few-shot	1.0000	ns
Qwen2.5-Instruct	qwen-3b	gsm8k	zero-shot vs CoT	0.0000	***
Qwen2.5-Instruct	qwen-3b	gsm8k	zero-shot vs structured	1.0000	ns
Qwen2.5-Instruct	qwen-3b	mmlu	zero-shot vs few-shot	1.0000	ns
Qwen2.5-Instruct	qwen-3b	mmlu	zero-shot vs CoT	0.0035	**
Qwen2.5-Instruct	qwen-3b	mmlu	zero-shot vs structured	1.0000	ns
Qwen2.5-Instruct	qwen-7b	sst2	zero-shot vs few-shot	0.0002	***
Qwen2.5-Instruct	qwen-7b	sst2	zero-shot vs CoT	0.0000	***
Qwen2.5-Instruct	qwen-7b	sst2	zero-shot vs structured	0.0000	***
Qwen2.5-Instruct	qwen-7b	gsm8k	zero-shot vs few-shot	0.0000	***
Qwen2.5-Instruct	qwen-7b	gsm8k	zero-shot vs CoT	0.0000	***
Qwen2.5-Instruct	qwen-7b	gsm8k	zero-shot vs structured	0.0000	***
Qwen2.5-Instruct	qwen-7b	mmlu	zero-shot vs few-shot	0.6774	ns

Qwen2.5-Instruct	qwen-7b	mmlu	zero-shot vs CoT	0.0000	***
Qwen2.5-Instruct	qwen-7b	mmlu	zero-shot vs structured	0.5123	ns
Llama-3	llama-1b	sst2	zero-shot vs few-shot	0.0000	***
Llama-3	llama-1b	sst2	zero-shot vs CoT	0.0653	ns
Llama-3	llama-1b	sst2	zero-shot vs structured	0.0002	***
Llama-3	llama-1b	gsm8k	zero-shot vs few-shot	0.0059	**
Llama-3	llama-1b	gsm8k	zero-shot vs CoT	1.0000	ns
Llama-3	llama-1b	gsm8k	zero-shot vs structured	1.0000	ns
Llama-3	llama-1b	mmlu	zero-shot vs few-shot	0.0000	***
Llama-3	llama-1b	mmlu	zero-shot vs CoT	0.0844	ns
Llama-3	llama-1b	mmlu	zero-shot vs structured	0.0001	***
Llama-3	llama-3b	sst2	zero-shot vs few-shot	0.0000	***
Llama-3	llama-3b	sst2	zero-shot vs CoT	0.0293	*
Llama-3	llama-3b	sst2	zero-shot vs structured	0.3074	ns
Llama-3	llama-3b	gsm8k	zero-shot vs few-shot	0.0000	***
Llama-3	llama-3b	gsm8k	zero-shot vs CoT	0.0000	***
Llama-3	llama-3b	gsm8k	zero-shot vs structured	0.5982	ns
Llama-3	llama-3b	mmlu	zero-shot vs few-shot	0.0000	***
Llama-3	llama-3b	mmlu	zero-shot vs CoT	0.2415	ns
Llama-3	llama-3b	mmlu	zero-shot vs structured	0.0482	*
Llama-3	llama-8b	sst2	zero-shot vs few-shot	0.0000	***
Llama-3	llama-8b	sst2	zero-shot vs CoT	0.4230	ns
Llama-3	llama-8b	sst2	zero-shot vs structured	1.0000	ns
Llama-3	llama-8b	gsm8k	zero-shot vs few-shot	0.0000	***
Llama-3	llama-8b	gsm8k	zero-shot vs CoT	0.0000	***
Llama-3	llama-8b	gsm8k	zero-shot vs structured	0.0000	***
Llama-3	llama-8b	mmlu	zero-shot vs few-shot	0.0023	**
Llama-3	llama-8b	mmlu	zero-shot vs CoT	0.7598	ns
Llama-3	llama-8b	mmlu	zero-shot vs structured	0.7598	ns
Gemma-2	gemma-2b	sst2	zero-shot vs few-shot	0.0000	***
Gemma-2	gemma-2b	sst2	zero-shot vs CoT	0.5224	ns
Gemma-2	gemma-2b	sst2	zero-shot vs structured	0.0072	**
Gemma-2	gemma-2b	gsm8k	zero-shot vs few-shot	0.0216	*
Gemma-2	gemma-2b	gsm8k	zero-shot vs CoT	0.0000	***
Gemma-2	gemma-2b	gsm8k	zero-shot vs structured	0.6464	ns
Gemma-2	gemma-2b	mmlu	zero-shot vs few-shot	0.3707	ns
Gemma-2	gemma-2b	mmlu	zero-shot vs CoT	0.0002	***

Gemma-2	gemma-2b	mmlu	zero-shot vs structured	0.4576	ns
Gemma-2	gemma-9b	sst2	zero-shot vs few-shot	1.0000	ns
Gemma-2	gemma-9b	sst2	zero-shot vs CoT	1.0000	ns
Gemma-2	gemma-9b	sst2	zero-shot vs structured	0.2113	ns
Gemma-2	gemma-9b	gsm8k	zero-shot vs few-shot	0.0000	***
Gemma-2	gemma-9b	gsm8k	zero-shot vs CoT	0.0000	***
Gemma-2	gemma-9b	gsm8k	zero-shot vs structured	0.0673	ns
Gemma-2	gemma-9b	mmlu	zero-shot vs few-shot	0.8085	ns
Gemma-2	gemma-9b	mmlu	zero-shot vs CoT	0.8085	ns
Gemma-2	gemma-9b	mmlu	zero-shot vs structured	0.4523	ns

Significance: *** $p < .001$, ** $p < .01$, * $p < .05$, ns = not significant.

Appendix D. Compute and reproducibility

All experiments ran on a single Apple M4 laptop (16 GB unified memory) via MLX, one model resident at a time. Models are the 4-bit `mlx-community` quantizations of Qwen2.5-Instruct (0.5/1.5/3/7B), Llama-3 (Llama-3.2-1B/3B, Meta-Llama-3.1-8B), and Gemma-2 (2B, 9B). Decoding is greedy (temperature 0). Max new tokens per condition: 8 (MMLU), 16 (SST-2), 48 (GSM8K non-CoT), 256 (SST-2/MMLU CoT), 512 (GSM8K CoT). $n = 300$ examples per condition; MMLU is stratified across all 57 subjects. Primary results use seed 42; a second seed (43) is reported for variance where available. The full harness (`run_experiments_local.py`), prompt builder, extraction code, prompt templates, and per-example results are released at github.com/ssamalsamir/cot-model-specificity — `--aggregate` reproduces all tables from the saved per-example CSVs.